SUB-MODULE 7.5 · DATA PREPARATION & EVALUATION

Fine-Tuning Evaluation Metrics

Perplexity, downstream benchmarks, LLM-as-judge, and human evaluation — a complete measurement toolkit

10 min	B4 – Analyse	L2 – Intermediate	AI-FT-IN-TH-000017
Duration	Bloom's	Level	ID

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The 4 Evaluation Layers

Layer 1: Perplexity

Measures model confidence on a held-out test set. Lower = better. Quick check during training. Does NOT correlate well with human preference.

Layer 2: Benchmarks

MMLU (knowledge), HumanEval (code), MT-Bench (instruction following), TruthfulQA (factuality). Automated — fast and reproducible.

Layer 3: LLM-as-Judge

Use GPT-4/Claude to score responses on a 1–5 rubric. Cheap, scalable, correlates well with human preference. Risks: judge biases.

Layer 4: Human Evaluation

A/B comparisons by human raters. Most trustworthy but expensive ($0.50–5.00/example). Required for final production decisions.

Key Benchmark Suite for Fine-Tuned LLMs

Benchmark	Task Type	Metric	Fine-Tune Relevant?
MMLU	57-subject knowledge test	5-shot accuracy	✅ Knowledge retention
HumanEval	Python code generation	pass@1 (exec. accuracy)	✅ Code fine-tuning
MT-Bench	Multi-turn instruction following	GPT-4 score 1–10	✅ Chat fine-tuning
TruthfulQA	Factual accuracy under pressure	% truthful answers	✅ Hallucination check
AlpacaEval	Open-ended instruction quality	Win rate vs GPT-4	✅ General fine-tuning
GSM8K	Grade-school math reasoning	CoT accuracy	✅ Reasoning tasks

LLM-as-Judge Setup — GPT-4 Evaluation Prompt

```
import openai
client = openai.OpenAI()

JUDGE_PROMPT = """You are an expert AI evaluator. Rate the following
response to the instruction.

Instruction: {instruction}
Response: {response}

Rate on these dimensions (1-5 each):
1. Correctness: Is the response factually accurate?
2. Helpfulness: Does it fully address the instruction?
3. Clarity: Is it well-written and easy to understand?
4. Conciseness: Is it appropriately concise (not padded or truncated)?

Return ONLY valid JSON: {{"correctness": N, "helpfulness": N, "clarity":
N, "conciseness": N, "overall": N}}"""

def llm_judge(instruction, response):
    result = client.chat.completions.create(
        model="gpt-4o",
        messages=[{"role":"user", "content": JUDGE_PROMPT.format(
            instruction=instruction, response=response)}],
        response_format={"type": "json_object"}
    )
    return json.loads(result.choices[0].message.content)

# Evaluate fine-tuned model vs base model
for ex in eval_set[:100]:
    ft_scores = llm_judge(ex["instruction"],
ft_model_output(ex["instruction"]))
    base_scores = llm_judge(ex["instruction"],
base_model_output(ex["instruction"]))
```

Key bias to watch: GPT-4 as judge has verbosity bias (prefers longer answers) and self-enhancement bias (prefers GPT-4-like outputs). Use multiple judge models.

KEY REFERENCES

- Zheng, L. et al. (2023). Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. NeurIPS 2023.
- Hendrycks, D. et al. (2020). Measuring Massive Multitask Language Understanding (MMLU). ICLR 2021.
- Chen, M. et al. (2021). Evaluating Large Language Models Trained on Code (HumanEval). arXiv.

▶ Next: Module 7 Assessment + Mini Project